॥वसुधैव कुटुम्बकम्॥
SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)
(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC
Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

A PROJECT REPORT

ON

Transformer-Based Depression Detection from Social Media Text Using Contextual Language Representations

A project report submitted in partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering

Submitted By
Harshini Bonde 24070521015
Vedant Nawghare 24070521047
Shravani Sadawarte 24070521287

UNDER THE GUIDANCE OF

Dr. Ruchi Patel

Associate Professor

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

AY 2026-27

॥वसुधैव कुटुम्बकम्॥
SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)
(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC
Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

.

CERTIFICATE

This is to certify that the Project work entitled **Transformer-Based Depression Detection from Social Media Text Using Contextual Language Representations** is carried out by **Harshini Bonde, Vedant Nawghare, Shravani Sadawarte**, in partial fulfillment for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering**, Symbiosis International (Deemed University), Pune during the academic year 2026-27.

Dr. Ruchi Patel
Associate Professor

Dr. Giridhar Urkude
Project In-charge

Dr. Sagarkumar S. Badhiye
Head of Department, CSE

Dr. Nitin Rakesh
Director, SIT Nagpur

DECLARATION

We hereby declare that the project titled **Transformer-Based Depression Detection from Social Media Text Using Contextual Language Representations** submitted to Symbiosis Institute of Technology, Constituent of Symbiosis International (Deemed University) Pune for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a result of original research carried out by us. We understand that our report may be made electronically available to the public. It is further declared that the project report or any part thereof has not been previously submitted to any University or Institute for the award of any degree or diploma.

Name(s) of Student(s): Harshini Bonde, Vedant Nawghare, Shravani Sadawarte

PRN: 24070521015, 24070521047, 24070521287

Degree: Bachelor of Technology in CSE

Department: Computer Science and Engineering

Title of the project: **Transformer-Based Depression Detection from Social Media Text Using Contextual Language Representations**

Harshini Bonde **Vedant Nawghare** **Shravani Sadawarte**

Date:

ACKNOWLEDGEMENT

We want to express our sincere gratitude to everyone who supported us throughout this project.

First and foremost, we would like to thank our project guide **Dr. Ruchi Patel**, for the valuable guidance, encouragement and feedback. This guidance has been a constant source of inspiration and motivation for us.

We are deeply thankful to **Prof. (Dr.) Nitin Rakesh, Director, Symbiosis Institute of Technology, Nagpur**, for providing the necessary facilities, resources and continuous support throughout this project.

We would also like to extend our gratitude to **Dr. Sagar Kumar Badhiye, Head of the Department (CSE)**, and **Dr. Giridhar Urkude, PBL Coordinator,** for their consistent guidance, encouragement, and valuable insights that greatly contributed to the successful completion of this work.

We want to acknowledge the contribution of our project team members, Harshini Bonde, Vedant Nawghare and Shravani Sadawarte, who have worked diligently and cooperated with us at every stage of the project. Their support and teamwork made this journey truly meaningful.

Lastly, we thank Symbiosis Institute of Technology, Nagpur, for giving us the opportunity to work on this project and to enhance our skills and knowledge.

ABSTRACT

Depression is a widespread mental-health condition, and social-media timelines offer a large, naturally occurring source of language and behavioural signals that can be studied computationally. Research has increasingly moved from single posts to user-level analysis, in which all posts of a user form one sample. System performance at this level depends not only on the model architecture but also on how the text is cleaned and represented, because aggressive preprocessing can remove expressive cues that are relevant to depression-related language.

This project report presents a controlled experimental study of the interaction between text representation and transformer architecture for user-level depression detection. Four representation conditions (Raw, Aggressive, Selective and Emotion-Based) are combined with seven transformer models (BERT, DistilBERT, Sentence-BERT, MentalBERT, EmoBERT, Longformer and BigBird), giving 28 configurations that all use 512-token inputs. Explicit first-person diagnostic-anchor posts are removed before the representation conditions are built, 53 user-level metadata features are fused with the text representation through gated multimodal fusion, and fixed, non-overlapping user-level training, validation and test partitions are used throughout. Performance is reported as accuracy, precision, recall, F1-score and ROC-AUC on the held-out test set.

Sentence-BERT with Raw representation achieved the highest F1-score (89.49%), with an accuracy of 92.96% and a ROC-AUC of 97.44%. MentalBERT with Raw representation followed closely (F1 89.28%, accuracy 92.83%), and Longformer with Selective representation achieved an F1-score of 88.72% and the highest ROC-AUC in the matrix (97.89%). No single representation was best for every architecture, and Aggressive Cleaning did not give the best F1-score for any of the seven architectures. Explainability and representation analyses using SHAP, LIME and t-SNE were conducted for selected model–condition configurations to complement the quantitative evaluation. These analyses provide illustrative views of metadata contribution, individual predictions and representation structure; they were not applied uniformly to all 28 configurations.

The findings are classification results on a benchmark dataset with self-disclosure-based labels. They are not clinical diagnoses, and because every configuration uses 512-token inputs, the study does not measure the effect of longer input length.

Keywords: Depression detection, Social media, Transformer models, Text representation, Text preprocessing, Metadata fusion, Emotion-based representation, Explainable AI

TABLE OF CONTENTS

Page No.

Certificate ..... **ii**

Declaration ..... **iii**

Acknowledgement ..... **iv**

Abstract ..... **v**

Table of Contents ..... **vii**

List of Tables ..... **ix**

List of Figures ..... **x**

List of Abbreviations ..... **xi**

CHAPTER 1: INTRODUCTION **1**

1.1 Introduction 1

1.2 Problem Statement 2

1.3 Scope of Research 2

1.4 Research Questions 2

1.5 Objectives 3

1.6 Organization of the Report 3

CHAPTER 2: LITERATURE REVIEW **5**

2.1 Literature Study 5

2.1.1 Social-Media Depression Detection and Benchmark Datasets 5

2.1.2 Feature-Based and Early Machine-Learning Approaches 5

2.1.3 Deep Learning and Hybrid Neural Architectures 6

2.1.4 Emotion-Aware Depression Detection 6

2.1.5 Transformer-Based Depression Detection 7

2.1.6 Text Preprocessing and Information Preservation 8

2.1.7 Context Length and Long-Context Modelling 8

2.1.8 Metadata and Multimodal Fusion 9

2.1.9 Interpretability and Cross-Context Generalisation 9

2.1.10 Summary of Literature 10

2.2 Research Gap 11

CHAPTER 3: RESEARCH DESIGN AND EXPERIMENTAL FRAMEWORK **13**

3.1 Overview of the Framework 13

3.2 Experimental Design 14

3.3 Experimental Controls and Reproducibility 15

3.4 Software and Hardware 16

CHAPTER 4: IMPLEMENTATION 17
4.1 Overview 17
4.2 Dataset and Fixed User-Level Partitioning 17
4.3 Diagnostic-Anchor Removal 18
4.4 Text Cleaning and Representation Conditions 18
4.4.1 Raw / Minimal Cleaning (C1) 19
4.4.2 Aggressive Cleaning (C2) 20
4.4.3 Selective Cleaning (C3) 20
4.4.4 Emotion-Based Representation (C4) 20
4.5 Metadata Feature Extraction 21
4.6 Transformer Architectures 22
4.7 Multimodal Representation and Gated Fusion 23
4.8 Training and Class-Balancing Procedure 23
4.9 Training Configuration 24
4.10 Tokenisation and Context Length 24
CHAPTER 5: TESTING AND VALIDATION 25
5.1 Validation Overview 25
5.2 Evaluation Metrics 25
5.3 Leakage and Reproducibility Controls 26
5.4 Comparative Analysis Plan 26
5.5 Statistical Analysis 26
5.6 Explainability and Representation Analysis Methods 27
5.6.1 SHAP 27
5.6.2 LIME 27
5.6.3 t-SNE 28
5.7 Conclusion 27
CHAPTER 6: RESULTS AND DISCUSSION 29
6.1 Overall Performance Across the 4 × 7 Matrix 29
6.2 Effect of Text Cleaning and Representation 32
6.3 Transformer Architecture Comparison 34
6.4 Long-Context Architectures at 512 Tokens 35
6.5 Training Dynamics and Error Analysis 35
6.6 Explainability and Representation Analysis 38
6.6.1 Representation Quality Through t-SNE 38
6.6.2 Metadata Contribution and SHAP Analysis 38

6.6.3 Local Explanation Using LIME 38
6.6.4 Explainable AI Coverage 39
6.7 Comparison with the Base Study 39
6.8 Discussion of the Research Questions 40
6.9 Discussion Synthesis 40
CHAPTER 7: CONCLUSION AND FUTURE WORK 42
7.1 Conclusion 42
7.2 Limitations 43
7.3 Future Work 43
7.4 Ethical Aspects 43
REFERENCES 44

LIST OF TABLES

Table 1: Summary of representative studies on social-media depression detection 10

Table 2: The 4 × 7 experimental matrix 15

Table 3: Fixed user-level data split used across all 28 experimental configurations 17

Table 4: Text cleaning and representation conditions 19

Table 5: Composition of the 53-feature metadata vector 21

Table 6: Performance of the 28 transformer–representation configurations on the held-out test set (%) 29

Table 7: Representation condition giving the best F1-score for each architecture 33

Table 8: Best configuration of the present study and of the base study 40

LIST OF FIGURES

Figure 1: Overall experimental framework for user-level depression detection 9

Figure 2: Accuracy, F1-score and ROC-AUC across all 28 configurations of the 4 × 7 experimental matrix, grouped by transformer architecture 30

Figure 3: F1-score of the seven transformer architectures under the four representation conditions 31

Figure 4: ROC-AUC of the seven transformer architectures under the four representation conditions 31

Figure 5: Heat-map of F1-score across the seven transformer architectures and four representation conditions 32

Figure 6: Training and validation loss of Sentence-BERT + Raw over 10 epochs 36

Figure 7: Confusion matrix of Sentence-BERT + Raw on the held-out test set 36

Figure 8: t-SNE visualisation of fused test representations for Sentence-BERT + Selective 37

Figure 9: SHAP feature-importance plot of Sentence-BERT + Selective 38

Figure 10: LIME explanation for a true-positive prediction by Sentence-BERT + Selective 39

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AC	Aggressive Cleaning
AUC	Area Under the Curve
BERT	Bidirectional Encoder Representations from Transformers
BPE	Byte-Pair Encoding
D1 / D2 / D3	Depressed / non-depressed / depression-candidate user groups
FN	False Negative
FP	False Positive
LDA	Latent Dirichlet Allocation
LIME	Local Interpretable Model-agnostic Explanations
LLM	Large Language Model
MLP	Multi-Layer Perceptron
NLP	Natural Language Processing
PLM	Pretrained Language Model
ROC	Receiver Operating Characteristic
SC	Selective Cleaning
SHAP	SHapley Additive exPlanations
t-SNE	t-distributed Stochastic Neighbor Embedding
TN	True Negative
TP	True Positive

Chapter 1
Introduction

1.1 Introduction

Depression is a mental health disorder involving persistent changes in emotions, thoughts, behaviors and functioning. Given the ubiquity and the social and clinical impact, there is ongoing interest in approaches that might assist in identification of patterns at scale related to depression. The social media is one such source of evidence, where social media users regularly share emotions, experiences, interactions, and daily activities, and the computational analysis of these contents has emerged as a research field for exploring mental-health signals and depression detection [1, 2].

One particular problem formulation is user-level depression detection, in which the unit of analysis is the individual user instead of a single post. Symptoms of depression can often be spread throughout the user's timeline, or may be in past posts. Many early studies demonstrated that a linguistic and behavioural representation of a user is useful and subsequently richer and more context-aware representations were investigated. Methodological implication of user-level formulation is that all users' posts should be in the same data partition, so that the information of one user is not contained in both training and evaluation data.

These transformer-based language models have greatly expanded the scope of context information that can be learned from social-media text. The general-purpose encoders like BERT [3] and its variants represent the dependence among words in the general language, while domain-specific and task-specific variants learn to represent the specialized language [2]. There are two challenges with social-media timelines, though. They are linguistically variable, including slang, abbreviations, hashtags, mentions and platform-specific artefacts, and some of these components are not used as noise, but as affective and/or stylistic information. Aggressive normalisation will process the input with as little filtering as possible, but may lose exactly this information, whereas minimal filtering will filter the input as much as possible, leaving the surface form of the text fairly intact.

However, transformer architectures vary with respect to the number of text tokens they can handle in a single sequence. Normally 512 tokens will be accepted, but the number of tokens in a full user history might be much larger. This work is based on the base study which has motivated this work, which compared the performance of Longformer and

BigBird on both short and long inputs [4]. Both architectures are included here, but all 28 configurations of the present study involve 512-token inputs; thus, the present results do not show how increasing the input length from 512 to 4096 tokens affects either architecture..

Another worry is diagnostic leakage. First person, explicit depression diagnosis statements in social-media texts are possible. If these types of statements persist in the input, a classifier can take advantage of them directly, as opposed to learning patterns over a user's timeline. In this work, the explicit first-person diagnostic-anchor posts are detected and filtered out prior to building the text representations, so that all configurations start with the same anchor-free position.

This report thus considers text representation, transformer architecture and user-level metadata in concert at a single user-level protocol. For the selected configurations, complementary explainability and representation analyses (SHAP, LIME and t-SNE) are performed while four representation conditions (Raw, Aggressive, Selective and Emotion-Based) are crossed with seven transformer architectures.

1.2 Problem Statement

Usually, text preprocessing and transformer architecture are studied separately, namely by comparing preprocessing strategies on the same architecture or by comparing architectures, on the same fixed preprocessing pipeline. So the usefulness of a representation is hard to distinguish between a property of the representation and a property of the architecture in which it is used. Further, it is possible for a diagnosis to be spilled when the user level depression detection is explicit, and it is necessary to have partitions with no user appearing more than once when evaluating at the user level. This report focuses on the evaluation, using a single controlled user-level protocol with diagnostic-anchor removal and metadata fusion, of the performance of different text representations on different transformer architectures for depression detection from timelines of social media posts.

1.3 Scope of Research

The research covers the following:

- User-level binary classification of depressed (D1) and non-depressed (D2) users from the dataset introduced by Shen et al. [1]; the weakly labelled D3 group is not used.
- A 4 × 7 experimental matrix of four text representation conditions and seven transformer architectures, giving 28 configurations, all using 512-token inputs.
- Removal of explicit first-person diagnostic-anchor posts, fixed user-level partitions, training-only class balancing and training-only metadata normalisation.
- Fusion of the transformer text representation with 53 user-level metadata features through gated multimodal fusion, with an additional emotion branch in the Emotion-Based condition.
- Evaluation using accuracy, precision, recall, F1-score and ROC-AUC, supported by SHAP, LIME and t-SNE analyses on selected configurations.

Evaluation based on accuracy, precision, recall, F1-score and ROC-AUC, with further analyses of selected configurations with SHAP, LIME and t-SNE. The research does not include an evaluation for longer input lengths (4096 tokens), cross-platform or cross-population evaluation, clinical validation, or the causal interpretation of the metadata features or any deployed screening application. The results are classification results on a benchmark dataset, and should not be interpreted as clinical diagnoses.

1.4 Research Questions

The study is organised around four research questions. They are stated as questions because the experiments are descriptive comparisons and no formal hypothesis tests are performed.

- RQ1: How does the choice of text cleaning and representation condition affect user-level depression detection?
- RQ2: How does transformer architecture affect depression-detection performance?
- RQ3: Does the effect of the text representation depend on the transformer architecture?
- RQ4: How do long-context architectures compare with conventional short-context architectures under the evaluated experimental settings?

1.5 Objectives

- To analyse the effect of alternative text cleaning and representation strategies on user-level depression detection.
- To compare general-purpose, embedding-oriented, domain-oriented, emotion-oriented and long-context-capable transformer architectures under a common protocol.
- To examine whether the effectiveness of a representation condition varies with the transformer architecture.
- To observe how long-context-capable architectures behave in the same experimental matrix, and to complement the quantitative results with explainability and representation analyses.

1.6 Organization of the Report

Chapter 2 reviews related work and states the research gap. Chapter 3 describes the research design and experimental framework. Chapter 4 describes the implementation: the dataset and partitions, anchor removal, representation conditions, metadata features, architectures, fusion and training. Chapter 5 describes the evaluation and validation procedures. Chapter 6 presents and discusses the results of the 28 configurations and the explainability analyses. Chapter 7 gives the conclusion, limitations, future work and ethical considerations, followed by the references.

Chapter 2
Literature Review

2.1 Literature Study

The computational methods of depression identification from social media platforms have been going through similar stages. Initial research used manual linguistic, behavioral, sentiment and topic features. More recent research used deep learning to learn from text content, or to integrate text with behavioural or auxiliary modalities. Recently, transformer-based models, long-context architectures, emotion-aware representations, multimodal fusion and explainability methods have been introduced. The directions relevant to this report are summarized below and in table 1 at the end of this section, representative studies.

2.1.1 Social-Media Depression Detection and Benchmark Datasets

Social-media data of course allows for linguistic and behavioural data to be processed for mental-health research. In an approach by Shen et al. [1], they proposed a user-level depression-detection framework using a multimodal dictionary learning and a Twitter dataset, which is based on several groups of textual and behavioural features of each user rather than analysing messages one by one. This type of work is an incentive for the user level setting in the base study [4] and that of the present report.

The eRisk initiative has redefined mental-health analysis as an early-risk prediction problem, with the goal also being to determine if there is enough evidence of risk from an online history [5]. The τ-SS3 classifier continued this development by providing dynamic n-gram modelling of text streams [6]. The user is the unit of observation, both in early-risk prediction and in user-level classification. The classification is important at the user level as it is crucial to assign each user's posts to the training, validation or test partition; the vocabulary, topics and behaviour of these posts are common to several users. This report thus employs a single non-overlapping partition on the user level for all 28 configurations.

2.1.2 Feature-Based and Early Machine-Learning Approaches

The early days of depression detection were reliant on a variety of engineered features that related to word usage, posting behaviour, engagement and user attributes. The results of these studies demonstrated the complementarity of behavioural information to lexical information [7], [8]. While language alone is not enough to combine several clinically-motivated feature groups, as in the multimodal framework of Shen et al. [1], later works used multiple linguistics, behaviour, sentiment and topic information [9]. The literature provides motivation for the use of user-level metadata in this work; the metadata is considered as a complementary source fused with the representation of the text. The metadata vector consists of 53 features describing behavioural activity, language, readability, emotion, psychological markers, writing style and topic distributions.

2.1.3 Deep Learning and Hybrid Neural Architectures

Deep Learning changed the game from handcrafted representations to learned representations. Text was encoded using convolutional and recurrent attention networks and evidence for the timeline was aggregated using a hierarchical attention network. For instance, hierarchical convolutional attention models present information on multiple layers of a user's history [10]. Local linguistic information has been modeled in the broader context provided for each user by hybrid recurrent and attention based architectures [11] and [12]. To handle long user history, depressionNet provided a summary of the posts and combined it with other modalities [13], in contrast to this work, which extends the length of the input window, depressionNet summarizes the user history for the purpose of classification. The results of these studies indicate that the context provided to a model could be important methodologically, both in terms of quantity and type.

2.1.4 Emotion-Aware Depression Detection

Sentiment data is important for social-media mental-health analysis. Emotion-aware methods emphasize the words that are emotionally salient or add emotion-related representation mechanisms. Sentiment-aware attention and reinforcement-learning-based token selection, for instance, were given emphasis in the emotion-based reinforcement attention network (ERAN) [14] to highlight emotionally salient text. This work encourages the consideration of affective information as a possible signal besides content information of the text.

The current study investigates emotion-related information in two different ways. The former is the Emotion-Based representation condition where eleven emotion probabilities from a pretrained emotion-classification model are normalised, converted to a 64-dimensional vector and provided to the fusion stage as a separate emotion branch, instead of being embedded into the text to be presented to the fusion stage. The second is EmoBERT: Emotion-oriented Transformer as Text Encoder. The two mechanisms work different parts of the pipeline

2.1.5 Transformer-Based Depression Detection

Pretrained transformer language models use self-attention to calculate the contextual relations among an input sequence. BERT popularized the bidirectional transformer-encoder architecture [15], RoBERTa made the pretraining process better [16] and DistilBERT created a smaller, faster distilled BERT [17]. The MentalBERT pretraining model is modified to handle forum and social media text with mental health related themes [18]. In addition to these, depression-related signals have been explored using RoBERTa shared-task systems [19] and multimodal transformer systems which introduce additional modalities [20].

For depression detection, HSNPL proposed graph learning and prompt learning for interpretable detection [21] and large language models were fine-tuned and evaluated for social-media depression detection [22]. There are various kinds of transformer architectures covered in these studies, though there are only a few studies that are able to compare architectures and change the input-representation in the same experimental paradigm. In the present study, all the mentioned models are evaluated for representation under four conditions: General-Purpose, Embedding, Domain-specific, Emotion and Long-context.All the above models are thus evaluated on the same four representation conditions so that general-purpose, embedding-oriented, domain-specific, emotion-oriented and long-context capable architectures can be compared within one framework.

2.1.6 Text Preprocessing and Information Preservation

In social-media classification, the structures and informal language used in user-generated text have platform-specific characteristics, whose effects are important for correct preprocessing decisions. Normalisation of a URL, mention, hashtag, punctuation, digits, and repeated characters is done using conventional pipelines. Not only will this minimize noise, but it can also eliminate affect, negation, emphasis and writing style. The impact of these decisions on downstream models has been explored in a number of related text classifications problems [23], [24], [25], [26].

The question is particularly pertinent to transformer-based analysis as it allows a contextual model to bridge the gap between words and expressions in the same context, and features that are traditionally considered noise can provide the information necessary to perform the task. A comparison between Aggressive Cleaning and Selective Cleaning was performed in the base study [4] and Selective Cleaning was shown to be preferred. This provides a methodological basis for the study of information preservation in a broader sense, as this report elucidates in four conditions: Raw, Aggressive, Selective and Emotion-Based.

2.1.7 Context Length and Long-Context Modelling

However, traditional BERT-like models can only process approximately 512 tokens, and this might only be part of a user's lengthy history. Responding to this, DepressionNet did a summary of its history [13]. An alternative is to use architectures that process longer sequences. Both Longformer (which eliminates full self-attention on long inputs with local sliding-window attention and global attention [27]) and BigBird (which eliminates full self-attention on long inputs with block, random and global attention [28]).

The base study [4] is a direct forerunner of the current study of context length for user-level depression detection. It tested 512-token transformer baseline models and 512 and 4096-token models of Longformer and BigBird. It achieved the highest accuracy of 90.36%, F1-score of 87.15% and AUROC of 0.9667 in its best configuration which comprised Longformer with 4096-token input, Selective Cleaning and behavioural metadata fusion. The following two large context architectures are also included in the

present study in a broader representation-by-model comparison, albeit with only 512-token inputs. The conclusion that can be drawn from its outcomes then is therefore not that there is an effect of increasing the number of tokens from 512 to 4096, but that there is not.

2.1.8 Metadata and Multimodal Fusion

The information provided by the user at the user level may provide evidence relevant to depression that is not part of the text. Combination of textual and behavioural features have been utilized in various ways, including behavioural and psycholinguistic features that describe posting activity, linguistic style, emotional indicators, readability and topics distribution, and multimodal hierarchical attention models [9]. The value of the metadata can vary based on the representation of the text and the transformer that is applied to it. The study focuses on the fusion of the 53 dimensional user-level metadata vector with the transformer representation, and the addition of the emotion branch in the Emotion Based condition.

2.1.9 Interpretability and Cross-Context Generalisation

In mental-health machine learning, interpretability is important because predictive performance does not provide any indication of which parts of a timeline or which metadata features are used to make a prediction. There are general methods to help explain predictions made by a model, such as the one called SHAP (Structured Hessian of the Decision Function) [29] and work that adds interpretability to the model architecture, such as learning psychologically meaningful concepts and relations [21, 30].

An associated issue is generalisation. A model that works well in one group of data or population might not be applicable to other groups of data or populations, or might perform differently for other groups of data or populations. Abdelkadir et al. tested depression-detection models using a seven-country dataset, with the locations of the tweets, and reported challenges in the populations of the Global South [31]. This emphasises the need for tightly controlled experimental conditions when comparing models. In the present study, fixed non-overlapping user-level partitions are used alongside complementary analyses, namely, SHAP, LIME and t-SNE. These controls do not by themselves provide generalisation across populations or platforms, they only prevent user overlap across partitions.

2.1.10 Summary of Literature

Table 1: Summary of representative studies on social-media depression detection

Study (Year)	Technique / Model	Dataset(s)	Relevance to this work
[4] (2026)	Transformer preprocessing and long-context benchmark	Twitter	Examined preprocessing, metadata fusion and 512/4096-token context directly
[3]	Feature engineering with GRU-attention and ensemble	Twitter, Reddit, Facebook	Multi-platform feature and contextual modelling
[21] (2025)	Heterogeneous subgraph network with prompt learning	Social-media text	Interpretability and user/feature-level relationships
[22] (2025)	Fine-tuned large language models	Social-media depression dataset	LLM fine-tuning for depression classification
[31] (2024)	Cross-cultural transfer evaluation	Seven-country Twitter test setting	Limits of cross-population generalisation
[10] (2024)	Hierarchical convolutional attention	Twitter	Modelling at word/tweet and user-timeline levels
[9] (2023)	Multimodal hierarchical attention	Social-media text and behavioural features	Fusion of learned textual and behavioural representations
[14] (2022)	Emotion-based reinforcement attention	Social-media text	Emotion information in attention and token selection
[13] (2021)	DepressionNet: summarisation with multimodal fusion	Twitter	Long-history processing through post-history summarisation

Study (Year)	Technique / Model	Dataset(s)	Relevance to this work
[6]	τ-SS3 dynamic n-gram classifier	eRisk text streams	Dynamic text-stream modelling for early risk detection
[1] (2017)	Multimodal dictionary learning	User-level Twitter	User-level framework and dataset lineage
[5] (2017)	eRisk early-risk framework	Online user histories	Early-risk prediction setting and evaluation framework

2.2 Research Gap

The literature reviewed above reveals the following methodological gaps, which this study addresses:

- **Preprocessing and architecture are studied separately.** In comparison, cleaning strategies are compared on one architecture or architectures are compared on one fixed pipeline; it is not clear whether there is a dependence on the model to which the representation is fed. This study employs a 4 × 7 experimental design with conditions: Raw, Aggressive, Selective, and Emotion-Based conditions, and models: BERT, DistilBERT, Sentence-BERT, MentalBERT, EmoBERT, Longformer and BigBird.
- **Limited input length of conventional transformers.** There have been suggestions of using summarisation-based compression [13] and long-context architectures [27, 28], and the base study [4] provides evidence for extended context in its own context. This study is a wider representation comparison between Longformer and BigBird, but at 512 tokens only, meaning there is no isolation of the effect of the change from 512 to 4096.
- **Diagnostic leakage.** Explicit diagnostic self-disclosures can be used as direct predictive leakage in user level classification and previous studies have emphasised careful labelling and evaluation [1, 5]. Explicit first-person diagnostic-anchor posts are identified and eliminated in this case before the four representation conditions are developed and all configurations begin with the same timelines without anchor posts.
- **Combining several known components.** Several lines of research have focused on individual aspects of behaviour metadata [1, 9], emotion-aware methods [14],

transformer-based contextual representations [15, 18] and long-context architectures [4, 27, 28] have been explored to a limited extent. In this work all four representations, seven architectures, 53 metadata features, fixed user-level partitions and explainability analyses are put in one framework.

Chapter 3

Research Design and Experimental Framework

3.1 Overview of the Framework

The study suggests a user-level framework for analyzing the interaction between the text representation strategies and transformer-based language models in the classification of social media timelines with the aim of detecting depression. An experimental matrix of 28 configurations of the four text representation conditions and seven transformer architectures is systematically combined. All the user-level partitions are the same for all the configurations, except the text representation and the transformer architecture. The process is as follows. The data set is split according to the user level first. The explicit diagnostic self-disclosure anchoring is then eliminated and 53 user level metadata features are extracted and normalised. For one of the four conditions of text representation, the text is encoded with one of the transformer models, and for the Emotion Based condition an emotion representation is integrated. Gated multimodal fusion is used to fuse the representations and the fusion result is evaluated on the held out test set. The procedure is repeated for the seven architectures and four conditions. At the configuration level, classification metrics are compared and then explainability and representation-analysis methods are applied to selected configurations. The overall framework is presented in figure 1.

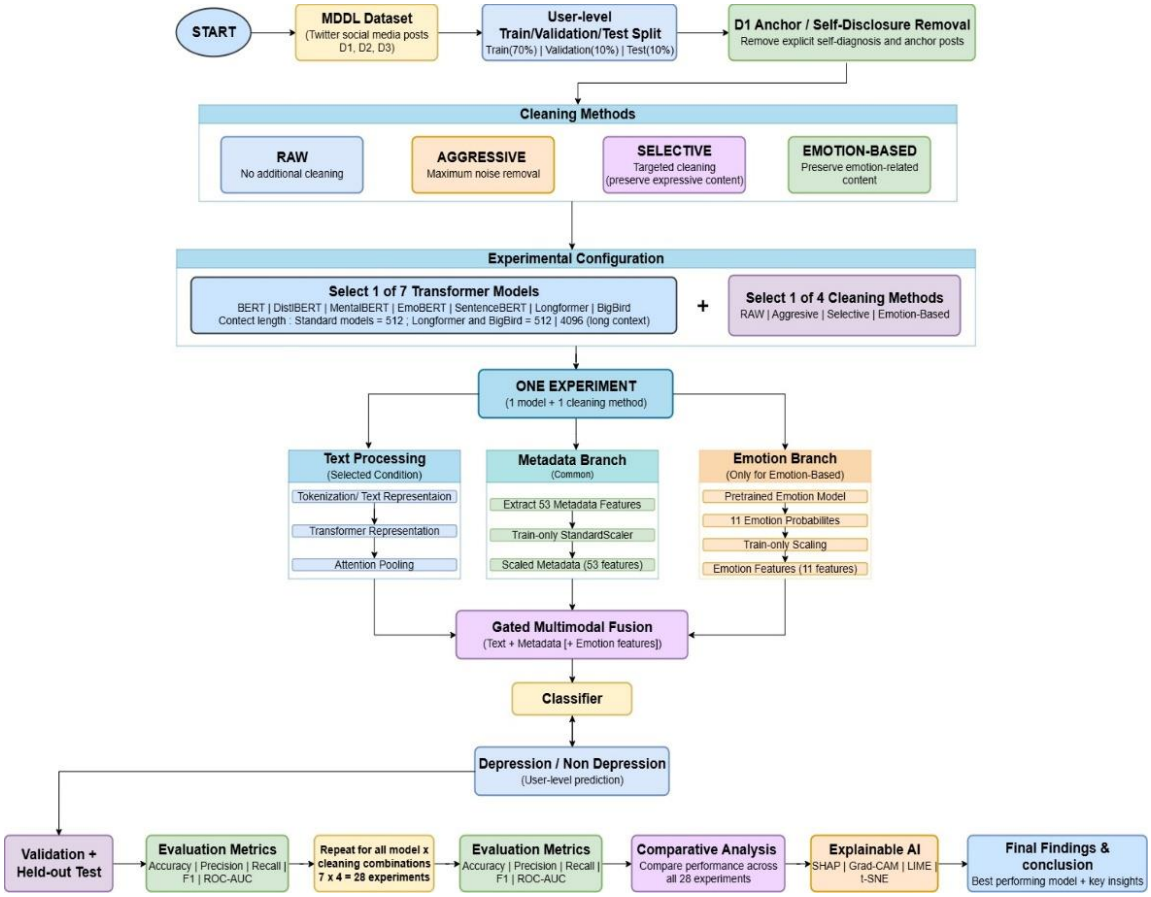


Figure 1: Overall experimental framework for user-level depression detection

3.2 Experimental Design

For the central design, there are 28 experimental runs (7 transformer designs × 4 text cleaning and representation conditions). All transformers are tested in all conditions that are listed in Table 2. All configurations use the same fixed user-level training, validation, and test partitions, diagnostic-anchor removal procedure, 53-feature metadata framework, training-set class-balancing strategy and evaluation protocol (unless otherwise indicated). Gated fusion is used to fuse the modalities available and the auxiliary emotion branch is only added in the Emotion-Based condition. Therefore, the experimental conditions are the text cleaning condition and the representation condition, and the transformer architecture.

Table 2: The 4 × 7 experimental matrix

Transformer architecture	Raw	Aggressive	Selective	Emotion-Based
BERT	✓	✓	✓	✓
DistilBERT	✓	✓	✓	✓
Sentence-BERT	✓	✓	✓	✓
MentalBERT	✓	✓	✓	✓
EmoBERT	✓	✓	✓	✓
Longformer	✓	✓	✓	✓
BigBird	✓	✓	✓	✓

The design tackles the four research questions presented in Section 1.4: the effect of the representation condition; differences between architectures; the dependence of the effect of the representation on the architecture; and how the evaluated architectures differ when presented with 512 tokens, including the sparse-attention architectures..

3.3 Experimental Controls and Reproducibility

Several controls are used in the pipeline to keep the comparison uniform and to reduce information leakage:

- The dataset is divided at the user level, so no user appears in more than one of the training, validation and test partitions.
- Diagnostic-anchor removal is carried out before the four representation conditions are created, so all four start from the same anchor-free timelines.
- Metadata normalisation is fitted on the training partition only and then applied to the validation and test partitions.
- Class balancing is applied to the training partition only; the validation and test partitions keep their original class distribution.
- All 28 configurations are evaluated on the same held-out test partition with the same metrics.

These safeguards reduce specific potential sources of leakage, but they do not by themselves guarantee that no other source of data or methodological bias remains.

3.4 Software and Hardware

The project leverages on the pre-trained transformer models and supporting components to represent text, extract emotions, and normalize the metadata for topic modelling. The models documented are: the cardiffnlp/twitter-roberta-base-emotion-latest model to get emotion probabilities, the all-mpnet-base-v2 checkpoint for Sentence-BERT, Latent Dirichlet Allocation (LDA) model for topic features, and StandardScaler to scale the metadata features. The software versions, hardware characteristics and full checkpoint names are not explicitly specified in the provided research paper and are therefore not reported here.

Chapter 4
Implementation

4.1 Overview

This chapter describes how the experimental framework was implemented: the dataset and its fixed partitions, diagnostic-anchor removal, the four text representation conditions, metadata feature extraction, the seven transformer architectures, gated multimodal fusion, class balancing, training and tokenisation. Each of the seven architectures has its own architecture diagram, and the common fusion design is explained in Section 4.7.

4.2 Dataset and Fixed User-Level Partitioning

The experiments are conducted on the user level social-media depression dataset as introduced by Shen et al. [1]. The concept of "unit of analysis" is user – all posts from a user are collected into a single user-level timeline. The classes used are D1 (users related to depression, positive class) and D2 (non-depressed users). Weakly labelled D3 group is not used in these experiments. There are 7,669 users in the experimental set, with 5,153 users in D2 and 2,516 in D1. There are three fixed, mutually exclusive partitions of users, which are summarised in Table 3. The partitioning is by user, so that, no post of one user is in more than one partition. The validation partition is employed in the model selection, and the test partition is reserved for the final performance. The same partitions are used for all 28 configurations.

Table 3: Fixed user-level data split used across all 28 experimental configurations

Partition	Total users	D2 (negative)	D1 (positive)
Train	6,135	4,122	2,013
Validation	767	515	252
Test	767	516	251
Total	7,669	5,153	2,516

The test counts of 516 D2 users and 251 D1 users are similar to the confusion matrix reported in Chapter 6, that is 483 + 33 + 21 + 230 = 767 users and 251 are positive. The final dataset has 7,669 users, with 6,135, 767, and 767 users in the training, validation, and test sets, respectively. There are 5,153 Users in the non-depression class (D2) and 2,516 Users in the depression related class (D1). Study documentation is incomplete as it does not record all of the intermediate filtering stages or the number of users and posts removed at each of the stages.

4.3 Diagnostic-Anchor Removal

Direct statements of a diagnosis can be very explicit lexical cues for the target class. To mitigate this diagnostic leakage, explicit diagnostic or self-disclosure messages (written in the first person) are tagged and removed prior to building the four representation conditions. This is important in user-level detection because an isolated such statement can practically contain the label, rather than the model having to extract depression-related patterns from the linguistic and behavioural evidence from the entire timeline.

The anchor detector identifies first person statements of diagnosis or disclosure of Depression including, “I have depression” and “I was diagnosed with depression”. No removal of anchors in clearly third person or quoted cases. Records that are detected are removed from D1 timelines, and are not modified in D2 timelines. The procedure identified 1,920 raw anchor records, 1,847 distinct (username, tweet_id) pairs and 73 duplicated links. The records in the training partition were deleted, 1535 records, in the validation partition 199 records were deleted and in the test partition 186 records were deleted. These anchor-free timelines are the starting points for all four representation conditions, and so it is not the amount of exposure to the anchors that causes differences between conditions.

4.4 Text Cleaning and Representation Conditions

The anchor-free timelines are subjected to four conditions that retain, remove or augment different kinds of information, summarised in Table 4.

Table 4: Text cleaning and representation conditions

Condition	Treatment of the anchor-free timeline
Raw / Minimal Cleaning (C1)	Only minimal changes; punctuation, emoji, hashtags, mentions, capitalisation, repeated characters and URLs are largely left untouched. Serves as the condition closest to unmodified social-media input.
Aggressive Cleaning (C2)	Strong normalisation: retweet symbols, URLs, mentions, hashtags, HTML entities, emoji, non-ASCII symbols, punctuation, digits and underscore artefacts are removed; repeated characters and spaces are reduced; text is lower-cased.
Selective Cleaning (C3)	Targeted normalisation that retains emotionally, psychologically or stylistically meaningful content such as emotional words, negations, repeated characters and sentiment-bearing words, while removing unnecessary surface-level or technical noise.
Emotion-Based (C4)	An auxiliary affective representation rather than a text-cleaning operation: 11 emotion probabilities are computed for each timeline and passed through a dedicated emotion branch to the fusion stage.

4.4.1 Raw / Minimal Cleaning (C1)

In the Raw condition, the anchor-free timeline is retained, with some minor modifications. Social-media characteristics such as punctuation, emoji, hashtags, mentions, capitalisation, repeated characters and URLs are mostly left as they are. It shows how much the transformer models can use in the case of the majority of the social media text used essentially unchanged.

4.4.2 Aggressive Cleaning (C2)

Aggressive Cleaning applies a strong amount of normalisation. Symbols, URLs, mentions, hashtags, HTML entities and emoji and non-ASCII symbols are stripped, redundant character repetitions and spaces are eliminated, and the text is converted to lower case. It's a good noise-reduction tactic, and it's the opposite of conditions that maintain expressive social-media information.

4.4.3 Selective Cleaning (C3)

Selective Cleaning is a process that normalizes the text while preserving information that is emotionally, psychologically or stylistically significant to the text. Surface level or technical noise is stripped away, and grammatically important and affective elements like emotional words, negations, repeated characters, and sentiment-bearing words are retained. As pointed out by previous studies on the classification of depression in social media [1, 3, 4], psychologically relevant features of texts, particular to the platforms, are informative for classification and the study of depression is following the selective-cleaning approach of the base long-context study.

4.4.4 Emotion-Based Representation (C4)

In the Emotion-Based condition there is an additional affective representation. All anchor-free timelines are sent to the pretrained model cardiffnlp/twitter-roberta-base-emotion-latest for scoring for 11 emotions: anger, anticipation, disgust, fear, joy, love, optimism, pessimism, sadness, surprise and trust. This is a 11-dimensional vector that can be used independently as an auxiliary channel. The scores are standardized using a scaler trained on the training partition, and then mapped onto a 64 dimensional emotion representation. The emotion are not merged into the token sequence of the transformer or blended in the

text embedding, but are provided separately at the fusion step. The Emotion-Based condition needs to be differentiated from the EmoBERT architecture. The condition is added to the downstream fusion framework as an auxiliary emotion representation, which can be fused with any text encoder and is a pretrained emotion-classification model. EmoBERT is an encoder that is based on the transformer. It is, therefore, a representation and fusion strategy as opposed to a text-encoding model as described by EmoBERT.

4.5 Metadata Feature Extraction

In addition to the transformer text representation, 53 user-level metadata features are extracted from each anchor-free user timeline. They describe user activity, linguistic characteristics, readability, emotional reactions, psychological markers, writing style and topic distributions. They are organised into seven groups, listed in Table 5.

Table 5: Composition of the 53-feature metadata vector

No.	Feature group	Count	Description
1	Behavioural	8	Posting frequency, temporal activity patterns and account-level behavioural characteristics derived from the timeline
2	Linguistic	4	Vocabulary richness and lexical-diversity measures
3	Readability	3	Sentence-structure complexity and standard readability indices on the aggregated text
4	Emotional	6	Lexicon-based emotional-intensity signals reflecting the affective tone of the timeline
5	Psychological	7	Psychologically oriented markers such as first-person pronoun use, negation frequency and affective word-category counts
6	Writing style	5	Punctuation usage, capitalisation tendencies and other author-level stylistic cues
7	LDA topics	20	Latent Dirichlet Allocation topic-distribution features representing each user's thematic content
	Total	**53**	

All 53 features are stored in one user-level metadata vector. Features are standardised with a StandardScaler that is fitted on the training partition (fit-transform) and applied unchanged to the validation and test partitions (transform), so validation and test data do not influence the normalisation parameters.

The available project documentation specifies the seven metadata feature groups and their respective counts, but does not provide a verified list of every individual feature name. The report therefore describes the metadata representation at the group level and does not infer undocumented feature names.

4.6 Transformer Architectures

Seven transformer-based architectures were evaluated in this study: BERT, DistilBERT, Sentence-BERT, MentalBERT, EmoBERTa, Longformer, and BigBird. These models were selected to compare general-purpose language representations, domain-oriented representations, sentence-level representations, and architectures designed to support longer input sequences.

Each architecture was evaluated under the same four text-representation conditions: Raw, Aggressive, Selective, and Emotion-Based. This produced a 4 × 7 experimental matrix containing 28 configurations. The comparison was conducted at the user level, with each sample representing a complete user timeline rather than an individual post.

The models were integrated into a common classification pipeline that combined the transformer-derived text representation with user-level metadata. The Emotion-Based condition additionally incorporated an auxiliary emotion representation. Model-specific tokenizers were used to prepare the text inputs.

Although Longformer and BigBird are designed to accommodate longer sequences, all configurations in the current experimental matrix used a maximum input length of 512 tokens. Therefore, the present comparison evaluates these architectures under a common input-length setting; it does not measure the benefit of increasing the input length.

The subsequent section describes the shared multimodal representation and fusion process used to combine the textual and auxiliary features.

4.7 Multimodal Representation and Gated Fusion

The classification framework combines textual representations with user-level metadata. The transformer encoder produces a representation of the user's text, while the metadata branch provides additional behavioral, linguistic, readability, emotional, psychological, writing-style, and topic-related information.

The metadata representation contains 53 features. These features are standardized using a StandardScaler fitted on the training data only. The fitted scaler is then applied to the validation and test sets to avoid using information from those sets during preprocessing.

For the Emotion-Based condition, an auxiliary emotion representation is incorporated in addition to the textual and metadata representations. This emotion representation is generated using the cardiffnlp/twitter-roberta-base-emotion-latest model and contains 11 emotion dimensions: Anger, Anticipation, Disgust, Fear, Joy, Love, Optimism, Pessimism, Sadness, Surprise, and Trust. The emotion features are processed separately before being combined with the other representations.

The fused representation is passed to the classification head to predict the user-level class. The two output classes correspond to D1, the depression-related class, and D2, the non-depression class. The same overall fusion strategy is used across the experimental configurations, while the textual representation depends on the selected transformer architecture and preprocessing condition.

The feature dimensions and classifier structure may vary by configuration; therefore, architecture-specific dimensions are reported only where they are explicitly documented in the implementation.

4.8 Training and Class-Balancing Procedure

The original training partition contains 4,122 D2 users and 2,013 D1 users. The imbalance is handled by random undersampling in the training partition only: all 2,013 D1 users are kept and 2,013 D2 users are randomly selected from the D2 training users, giving a balanced training set of 4,026 users, half from each class. No undersampling or other rebalancing is applied to the validation and test partitions, which keep their original distributions (validation: 252 D1 and 515 D2 users; test: 251 D1 and 516 D2 users) for

model selection and final results. Balancing is done before model training, only on the training partition, and no users from the validation or test partitions are added to the balanced training set.

4.9 Training Configuration

Each model–condition combination is trained independently with a shared training framework: the same fixed user-level partitions, class-balancing strategy, metadata-normalisation protocol and evaluation procedure for all 28 configurations. The checkpoint for each configuration is the epoch with the highest F1-score on the validation set. The training and validation loss for the best configuration, Sentence-BERT + Raw, are plotted over 10 epochs in Chapter 6.

The supplied documentation does not specify the complete set of training hyperparameters, including the optimiser, learning rate, batch size, weight decay, learning-rate scheduler and random seed. These settings should be reported from the original training configuration files if they are available. Until verified, they are not assumed or inferred from the base study.

4.10 Tokenisation and Context Length

After the representation condition is applied, each user timeline is tokenised with the tokenizer of the selected transformer. Standard configurations of these models cap the input at about 512 tokens, and the 28 experiments use a 512-token input for all seven architectures. Longformer and BigBird use sparse attention to handle longer sequences [27], [28], but in this study they are evaluated on a smaller part of their context capacity, as the other models. The results therefore compare the evaluated architectures at one input length and say nothing quantitative about the gain from raising the input to 4096 tokens. The 4096-token results of the base study [4] come from a separate study and are treated as prior work rather than results of the present experiments.

Chapter 5
Testing and Validation

5.1 Validation Overview

The validation of this research is experimental. It consists of a uniform evaluation protocol applied to all 28 configurations, controls against data leakage, and complementary interpretability analyses. The experiments are classification experiments on a benchmark dataset; no clinical validation or real-world deployment is involved.

5.2 Evaluation Metrics

Performance is evaluated with metrics that stay informative under class imbalance. All reported results are computed on the held-out test partition, with D1 (depression) as the positive class. With TP, FP, TN and FN denoting true positives, false positives, true negatives and false negatives, the metrics are defined as follows.

- **Accuracy** is the overall proportion of correctly classified users:

  (TP + TN) / (TP + TN + FP + FN).

- **Precision** is the proportion of users predicted as D1 who are genuinely D1:

  TP / (TP + FP).

- **Recall** is the proportion of genuinely D1 users who are correctly identified:

  TP / (TP + FN).

- **F1-score** is the harmonic mean of precision and recall:

  2 × Precision × Recall / (Precision + Recall).

- **ROC-AUC** is the area under the receiver operating characteristic curve, obtained by integrating the true positive rate against the false positive rate over all decision thresholds applied to the predicted probability.

The test partition is imbalanced (251 D1 and 516 D2 users). A classifier predicting every test user as D2 would obtain an accuracy of about 67.3% but zero recall and zero F1-score for D1. F1-score is therefore the primary measure, and accuracy is interpreted together with precision, recall, F1-score and ROC-AUC. A fixed decision threshold of 0.5 is applied to the predicted class probabilities in all 28 configurations, which avoids adding

threshold tuning as another source of variation between configurations. The model checkpoint for each configuration is selected by the highest validation F1-score.

5.3 Leakage and Reproducibility Controls

The validation protocol comprises of the following elements: user-level partitions with no user overlap; diagnostics-anchor removal prior to building the top-level emotion and representation conditions; training-only fitting of the emotion-score scaler; training-only fitting of the emotion-scales scaler; training-only class balancing; and testing of all of the above test partitions using the same configuration. These controls enable a uniform comparison of the evaluated configurations.

5.4 Comparative Analysis Plan

The output from the 28 experiments, rather than being presented individually, is presented as an organised model-by-condition matrix. The analysis consists of the effects of the four representation conditions within each transformer, the difference between the seven architectures at matched conditions, and whether the effects of the representation depends upon the architecture, as well as comparing conventional transformer with sparse-attention architectures at 512 tokens. These reflect the four research questions of Section 1.4.

5.5 Statistical Analysis

The main results are descriptive comparisons of test-set performance across the 28 configurations. No inferential statistical test was performed, and the report makes no claim of statistical significance. Differences between configurations are reported as observed values from the experiments carried out.

5.6 Explainability and Representation Analysis Methods

The explainability and representation analyses were created for selected configurations and complement the aggregate metrics. They present complementary perspectives of feature contribution, individual predictions, and learned representations; no indication that all 28 configurations were analysed in the same way.

5.6.1 SHAP

SHAP [29] is based on Shapley values from cooperative game theory to attribute a model's predictions to the input features. In the multimodal model it identifies the metadata features (behavioural, linguistic, readability, emotional, psychological, writing-style and topic features) that are related to predictions. The SHAP feature-importance plot reported here is for Sentence-BERT with the Selective condition.

5.6.2 LIME

LIME provides an explanation for a single prediction, indicating the elements in the input space that are linked to a specific output of the model. A LIME explanation is specific to the chosen example and does not represent the importance of features on the entire test set as in the case of a global importance analysis. The explanation reported here is for a true-positive test example of Sentence-BERT with the Selective condition.

5.6.3 t-SNE

t-SNE maps learned representations into 2D space to be qualitatively examined and analyzed by the spatial arrangement of D1 and D2 test users. It is not used in training or used to select checkpoints for visualization. The plot reported here is for Sentence-BERT with the Selective condition. All 28 configurations are identical in terms of the 512-token input, so there is no comparison of short/long context.

For reasons of providing a verified implementation description and a corresponding result figure in the supplied project documentation, it is not reported if the gradient-based token-sensitivity or Grad-CAM type analysis was used to gain an understanding of the model. Therefore, only the documented SHAP, LIME and t-SNE analyses are discussed in the explainability discussion.

5.7 Conclusion

The evaluation applies the same metrics, thresholds, partitions and checkpoint-selection rule to all 28 configurations. This allows a fair descriptive comparison of representation conditions and architectures, with the limits stated in Section 5.5 and in Chapter 7.

Chapter 6

Results and Discussion

6.1 Overall Performance Across the 4 × 7 Matrix

The framework was tested in 28 controlled experiments obtained by combining seven transformer architectures with four text cleaning and representation conditions. All configurations were evaluated on the same held-out test partition, which contains 516 D2 and 251 D1 users, with accuracy, precision, recall, F1-score and ROC-AUC. F1-score is the main measure for comparing configurations, and accuracy and ROC-AUC give complementary information. Table 6 lists all results, and Figures 9, 10, 11 and 12 show them graphically.

Table 6: Performance of the 28 transformer–representation configurations on the held-out test set (%)

Transformer	Condition	Accuracy	Precision	Recall	F1-score	ROC-AUC
DistilBERT	Raw	87.74	80.08	83.27	81.64	94.60
	Aggressive	86.44	73.94	90.44	81.36	94.40
	Selective	81.88	67.50	86.06	75.66	92.31
	Emotion-Based	89.05	80.81	87.25	83.91	95.66
EmoBERTa	Raw	88.79	80.44	86.85	83.52	95.38
	Aggressive	84.75	74.28	81.67	77.80	93.01
	Selective	83.57	70.10	86.85	77.58	92.66
	Emotion-Based	86.70	76.14	86.45	80.97	94.91
MentalBERT	Raw	92.83	89.80	87.65	89.28	97.68
	Aggressive	86.57	73.57	92.03	81.77	95.59
	Selective	87.48	74.29	94.42	83.16	96.71
	Emotion-Based	90.48	81.56	91.63	86.30	96.57
BERT	Raw	89.05	77.93	92.83	84.73	96.91
	Aggressive	81.75	65.81	92.03	76.74	94.04
	Selective	85.79	72.61	90.84	80.71	94.88
	Emotion-Based	89.96	80.21	92.03	85.71	96.96
Sentence-BERT	Raw	92.96	87.45	91.63	89.49	97.44
	Aggressive	90.09	83.02	87.65	85.27	96.78
	Selective	90.22	82.59	88.84	85.60	96.50

Transformer	Condition	Accuracy	Precision	Recall	F1-score	ROC-AUC
	Emotion-Based	91.00	85.55	87.25	86.39	96.40
Longformer	Raw	89.05	79.51	89.64	84.27	96.17
	Aggressive	81.23	64.99	92.43	76.32	93.72
	Selective	92.31	85.29	92.43	88.72	97.89
	Emotion-Based	89.83	83.40	86.06	84.71	95.62
BigBird	Raw	88.14	74.84	96.02	84.12	96.49
	Aggressive	84.35	69.79	92.03	79.38	94.22
	Selective	83.05	67.44	93.23	78.26	94.21
	Emotion-Based	88.01	76.07	92.43	83.45	96.62

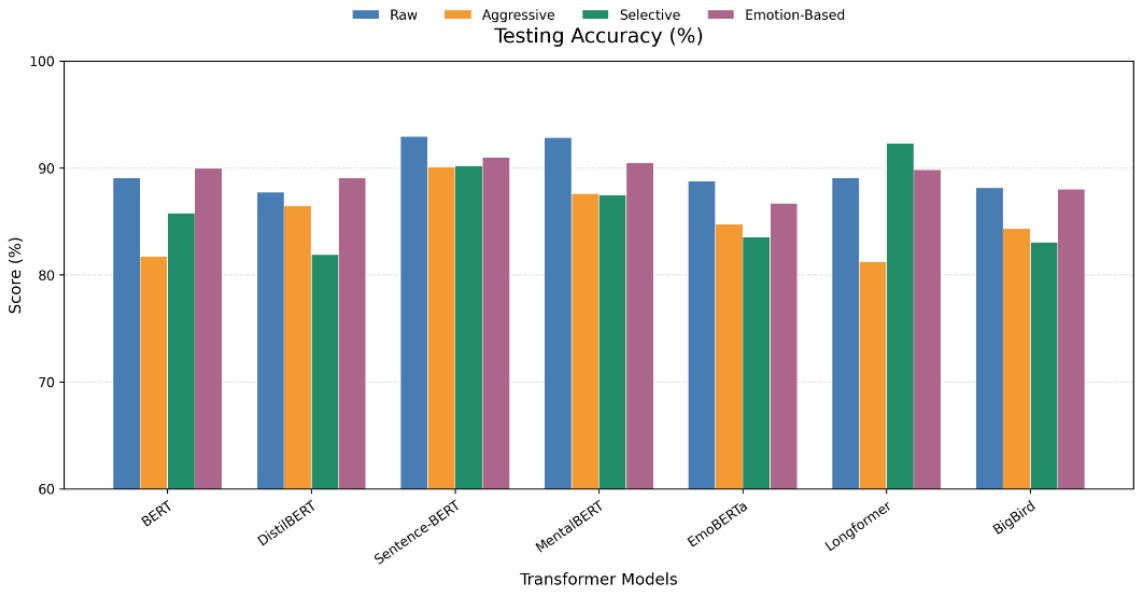


Figure 2: Accuracy, F1-score and ROC-AUC across all 28 configurations of the 4 × 7 experimental matrix, grouped by transformer architecture

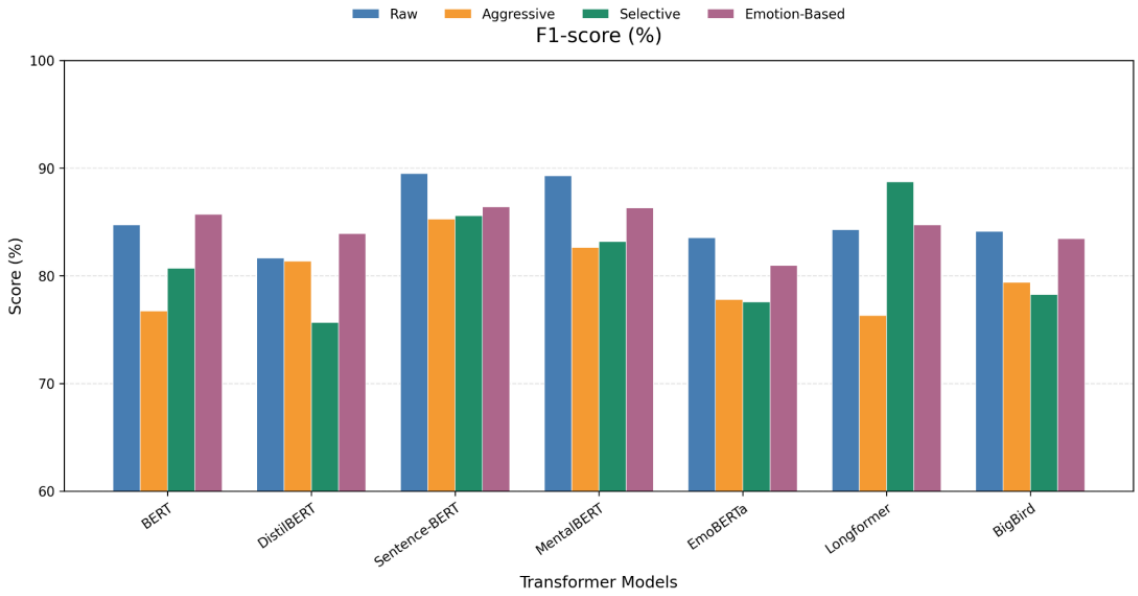


Figure 3: F1-score of the seven transformer architectures under the four representation conditions

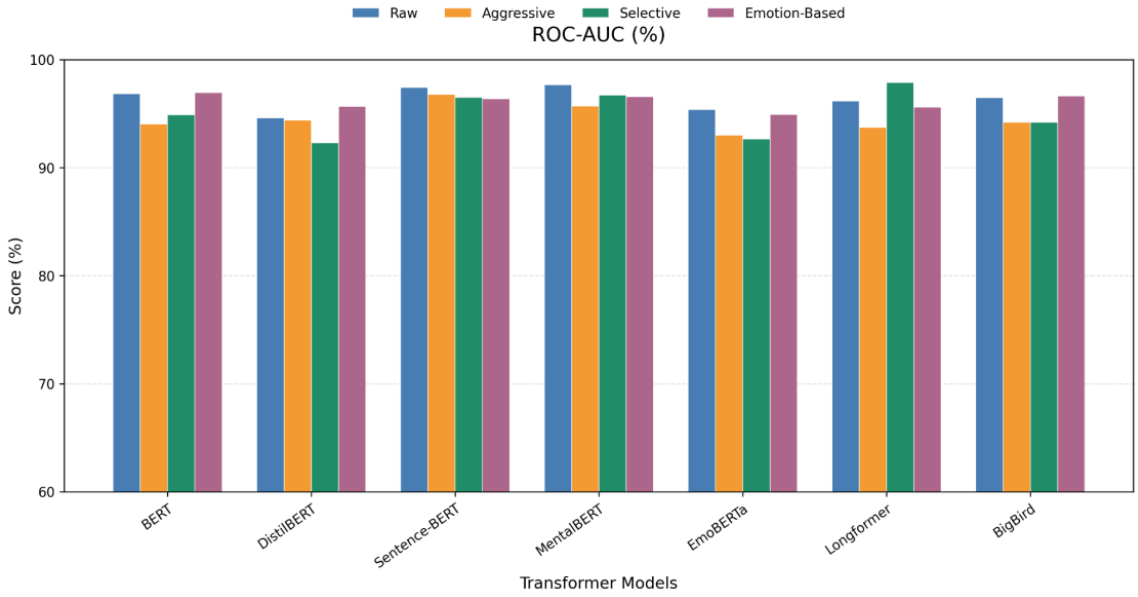


Figure 4: ROC-AUC of the seven transformer architectures under the four representation conditions

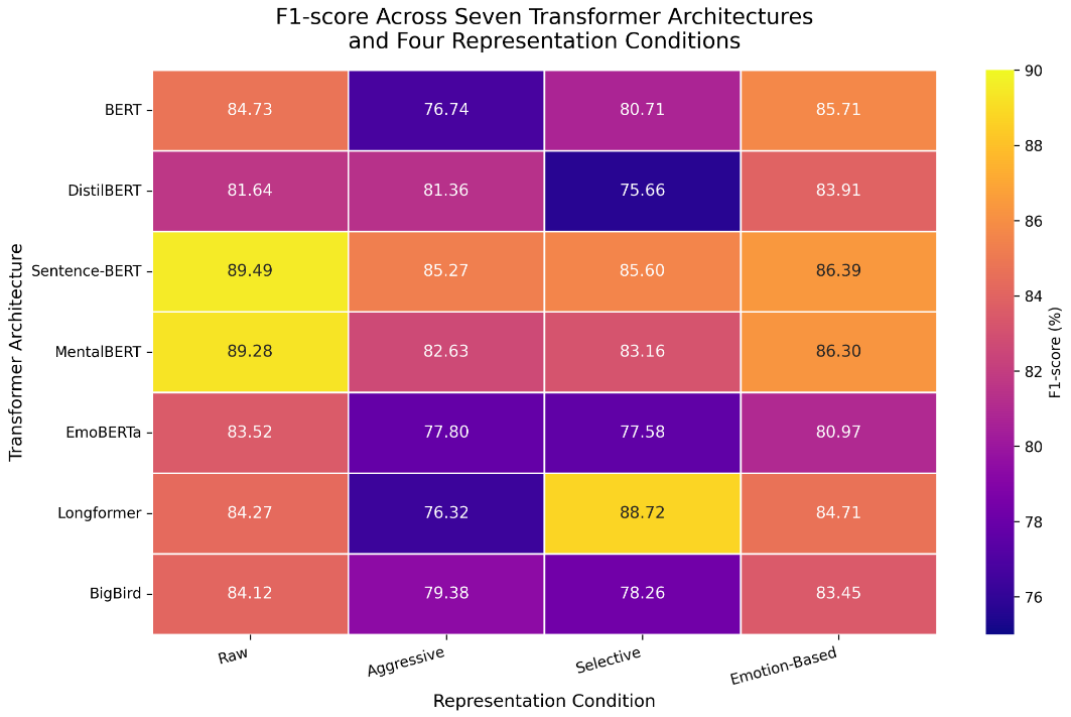


Figure 5: Heat-map of F1-score across the seven transformer architectures and four representation conditions

Sentence-BERT with Raw representation obtained the highest F1-score (89.49%), the highest accuracy (92.96%) and a ROC-AUC of 97.44%. MentalBERT with Raw representation was second on F1-score (89.28%) with an accuracy of 92.83% and a ROC-AUC of 97.68%. Longformer with Selective representation obtained an F1-score of 88.72%, an accuracy of 92.31% and the highest ROC-AUC in the matrix (97.89%). The best F1-score and the best ROC-AUC therefore do not belong to the same configuration: F1-score depends on the precision–recall balance at the fixed threshold of 0.5, while ROC-AUC measures discrimination across all thresholds.

At the other end, DistilBERT with Selective representation had the lowest F1-score (75.66%), followed by Longformer with Aggressive representation (76.32%) and BERT with Aggressive representation (76.74%). The range between the highest and lowest F1-scores is 13.83 percentage points, so performance is clearly not uniform across the model–condition combinations.

6.2 Effect of Text Cleaning and Representation

Cleaning and representation do not have the same impact for the seven transformers. The best F1-score is reached under different conditions for different architectures, as Table 7

shows. Raw representation gave the best F1-score for EmoBERT, MentalBERT, Sentence-BERT and BigBird; Emotion-Based representation for BERT and DistilBERT; and Selective representation for Longformer. Sentence-BERT achieved the best F1-score of 89.49% with Raw representation and MentalBERT achieved the best F1-score of 89.28% with Raw representation.

Table 7: Representation condition giving the best F1-score for each architecture

Transformer	Best condition	F1-score (%)	Aggressive F1-score (%)
BERT	Emotion-Based	85.71	76.74
DistilBERT	Emotion-Based	83.91	81.36
Sentence-BERT	Raw	89.49	85.27
MentalBERT	Raw	89.28	81.77
EmoBERT	Raw	83.52	77.80
Longformer	Selective	88.72	76.32
BigBird	Raw	84.12	79.38

No matter which of the 7 architectures was chosen, Aggressive Cleaning was not the top performer in terms of F1-score. Its F1-scores were 81.36% (DistilBERT), 77.80% (EmoBERT), 81.77% (MentalBERT), 76.74% (BERT), 85.27% (Sentence-BERT), 76.32% (Longformer) and 79.38% (BigBird). Selective Cleaning also outperforms Aggressive Cleaning for four of the seven architectures (BERT, MentalBERT, Sentence-BERT and Longformer) but not in the case of DistilBERT, EmoBERT or BigBird, meaning selective cleaning gives an advantage over aggressive cleaning in an architecture-dependent fashion.

The mean F1 score over the seven architectures for each of these models is computed as a purely descriptive summary from Table 6 and is 85.29% for Raw, 84.49% for Emotion-Based, 81.38% for Selective and 79.81% for Aggressive. The averages are provided for summarization of the matrix but not as outcome of a statistical test and mask the differences described above regarding the architectures.

The results indicate that the text representation text-classification model match is architecture-dependent. The high performance of the Raw representation demonstrates that the social media features of the text could be complemented by transformer

representation learning and in the case of BERT and DistilBERT proved to be beneficial as additional information. Several types of information are removed at once by Aggressive Cleaning; the experiments are not designed to isolate the type of information, so the reason for the differences observed cannot be assigned to the removal of any particular type of information.

6.3 Transformer Architecture Comparison

All seven architectures will have varying performance characteristics for all four conditions. Sentence-BERT (89.49%), MentalBERT (89.28%) and Longformer (88.72%) scored the best F1 score per architecture. Sentence-BERT also achieves the highest accuracy on the matrix (92.96%, Raw) and an F1-score of 85.27% to 89.49% on the different conditions, which shows that it performs relatively well. The values of MentalBERT's span from 81.77% (Aggressive) to 89.28% (Raw), which is larger than the range for the other models. While the best F1-scores of each architecture were found under the Emotion-Based condition (BERT: 85.71%, DistilBERT: 83.91%), the benefit of the auxiliary emotion representation differs based on architecture. EmoBERT achieved an F1-score of 83.52% and BigBird achieved 84.12% under Raw representation. Longformer achieved a higher accuracy (92.31%), F1 (88.72%) and ROC-AUC (97.89%) under Selective representation than the other. The observations indicate that the architecture and the representation of text need to be treated simultaneously: a good architecture for one representation does not necessarily maintain the same relative position in another.

6.4 Long-Context Architectures at 512 Tokens

The two sparse-attention architectures they support are Longformer and BigBird.They support two sparse-attention architectures: Longformer and BigBird. The best performance in the matrix was achieved by Longformer in the Selective representation category (F1-score 88.72%, ROC-AUC 97.89%). The best F1 score obtained was under Raw representation (84.12%, ROC-AUC 96.49%). But sparse-attention architectures are not necessarily superior to the other transformers examined here; Sentence-BERT + Raw (89.49%) and MentalBERT + Raw (89.28%) have higher F1-scores than Longformer + Selective (88.72%).

It is imperative to read in conjunction with the base study [4] which yielded best performance with Longformer, Selective Cleaning, metadata fusion and a 4096-token input (90.36% accuracy, 87.15% F1-score, 0.9667 AUROC). The values are not found in the 28 run matrix. The 512-token inputs are applied for all 7 of the architectures in this set, but the effect of a longer context was not directly tested. If there is a claim regarding the benefit of longer context window, a controlled comparison of input lengths with the same context lengths should be done under the same conditions.

6.5 Training Dynamics and Error Analysis

Sentence-BERT using Raw representation had the best F1 score. Figure 13 illustrates the training and validation loss for Sentence-BERT using Raw representation over 10 epochs. The training loss is monotonically decreasing in epochs. The validation loss decreases rapidly in the first epochs and then plateaus around epoch 4 after which it increases slightly while the training loss continues to decrease, causing the two curves to diverge. It is compatible with choosing the checkpoint based on the validation F1-score instead of taking the last epoch.

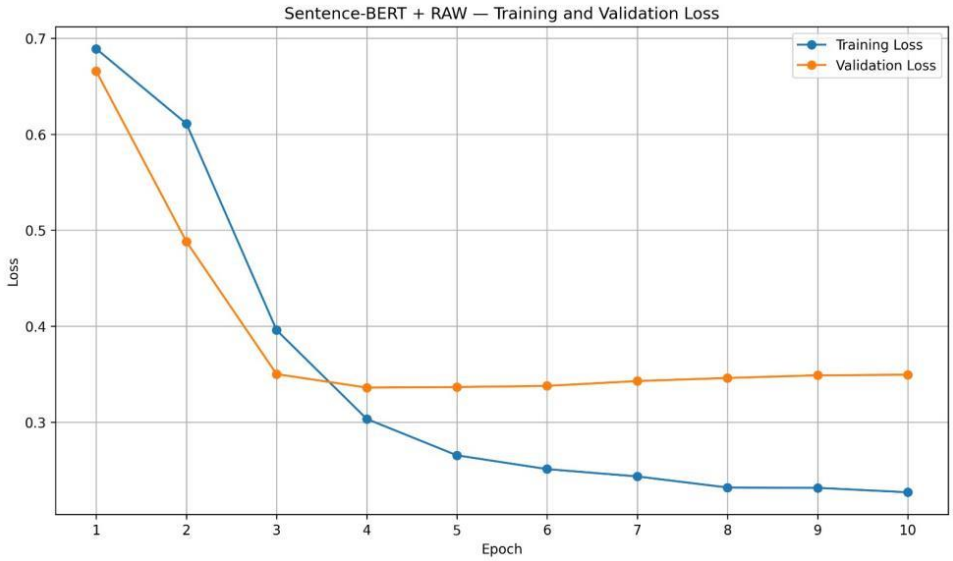


Figure 6: Training and validation loss of Sentence-BERT + Raw over 10 epochs

Figure 14 shows the confusion matrix of this configuration on the held-out test set: 483 true negatives, 33 false positives, 21 false negatives and 230 true positives, which add up to the 767 test users. These counts correspond to the reported accuracy of 92.96%, precision of 87.45%, recall of 91.63% and F1-score of 89.49%.

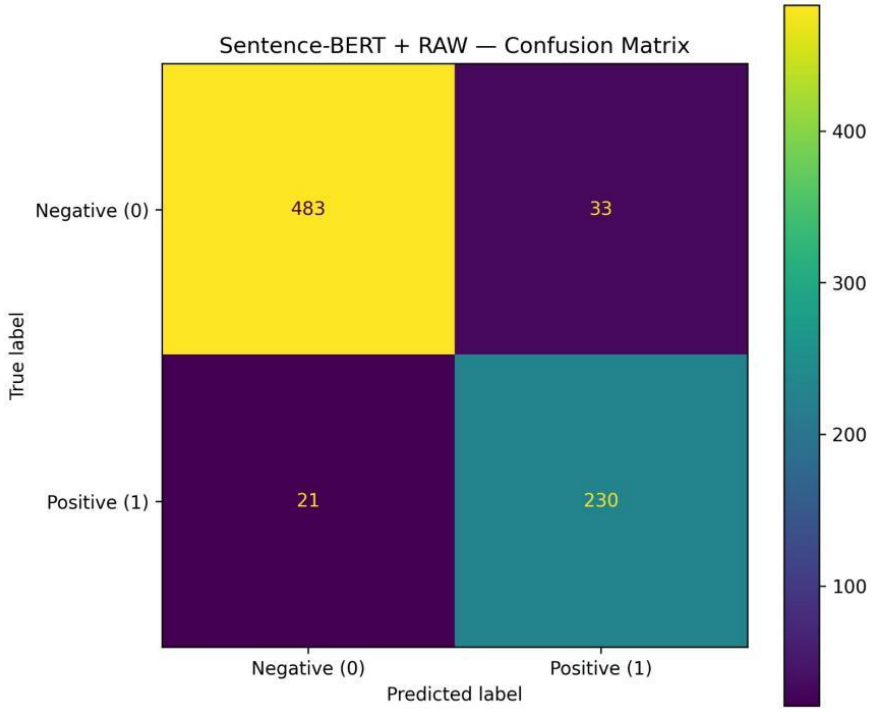


Figure 7: Confusion matrix of Sentence-BERT + Raw on the held-out test set

The balance of precision and recall also differs between configurations. BigBird + Raw has the highest recall in the matrix (96.02%) but a lower precision (74.84%), giving an F1-

score of 84.12%; it detects many D1 users but also produces many false positives. Longformer + Aggressive has a recall of 92.43% with a precision of only 64.99%, giving an F1-score of 76.32%. High recall therefore does not imply balanced classification, and with an imbalanced test partition precision, recall and F1-score must be read together.

6.6 Explainability and Representation Analysis

6.6.1 Representation Quality Through t-SNE

Figure 15 shows a t-SNE projection of the fused test representations of Sentence-BERT with Selective representation. The plot is a qualitative view of how D1 and D2 users are distributed in the learned space. In the projection one group of depressed users forms a visible cluster in the left-hand part of the plot, while most other depressed users overlap with the control users in the centre. t-SNE is a visualisation only; classification performance is judged by the quantitative metrics.

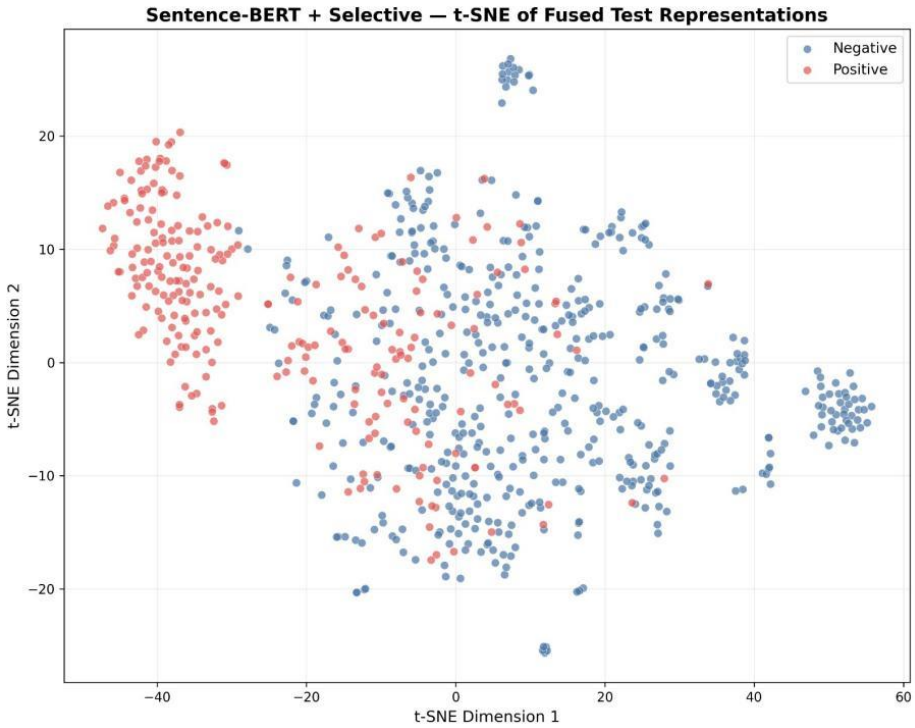


Figure 8: t-SNE visualisation of fused test representations for Sentence-BERT + Selective

6.6.2 Metadata Contribution and SHAP Analysis

The metadata features of Sentence-BERT with Selective representation's SHAP feature-importance plot is presented in figure 16. In this configuration, features towards the higher end of the plot (lda_topic_2, word_count, engagement, repeated_letters, lda_topic_9) have a wider range of SHAP values and thus a greater impact on the model's prediction. SHAP

values are expressions of how much a feature is contributing to the prediction of this model, not indications of clinical importance or causation of depression. The plot illustrates the role of metadata in the extension to the text representation, which complements the treatment of metadata as user level information in the base study [4].

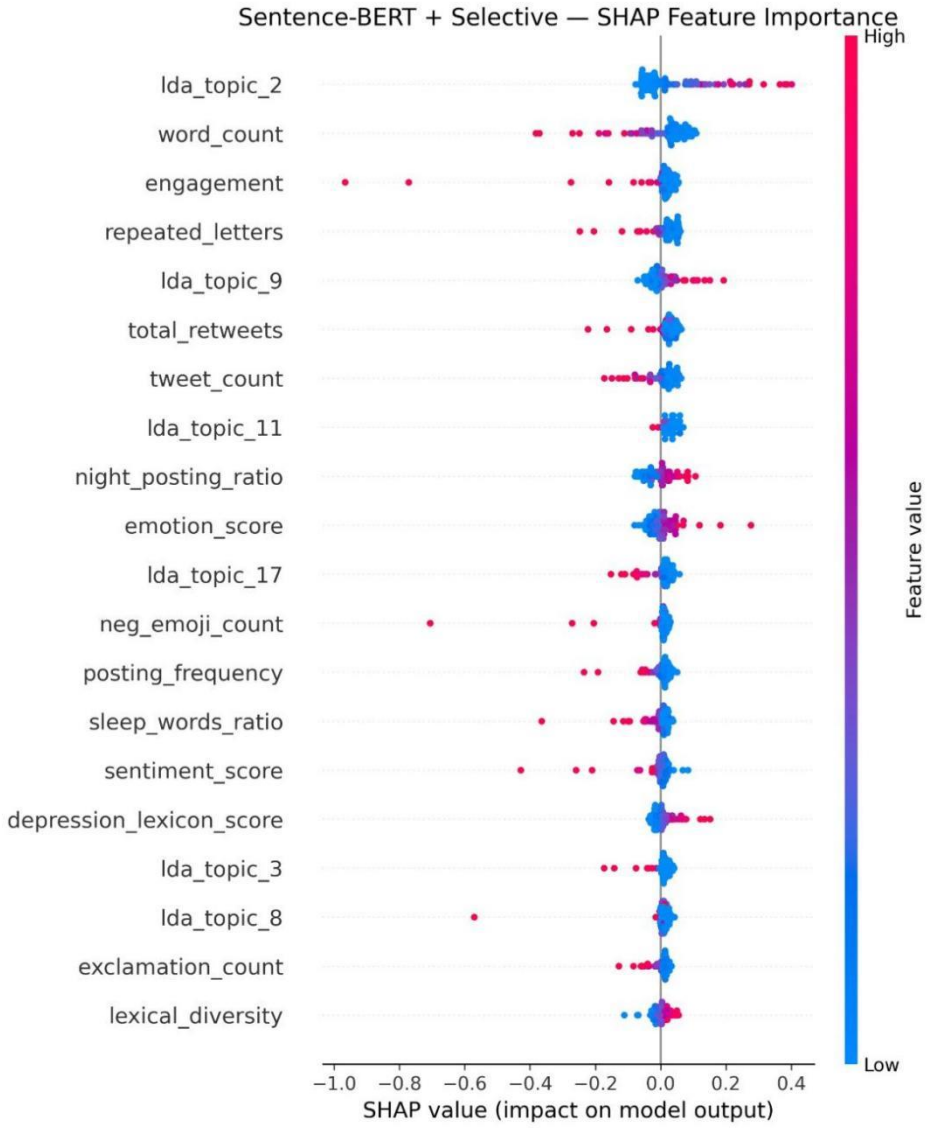


Figure 9: SHAP feature-importance plot of Sentence-BERT + Selective

6.6.3 Local Explanation Using LIME

Figure 17 shows a LIME explanation for a true-positive test prediction by Sentence-BERT with Selective representation. Bars to the right show tokens that increase the model's support for the depressed class for this example, for instance the token "i", "someone" and "texting", and bars to the left show tokens that reduce it, for example "you", "meant" and "2016". The explanation applies only to this example. It should not be generalised to all users or configurations, and it is not a clinical diagnosis.

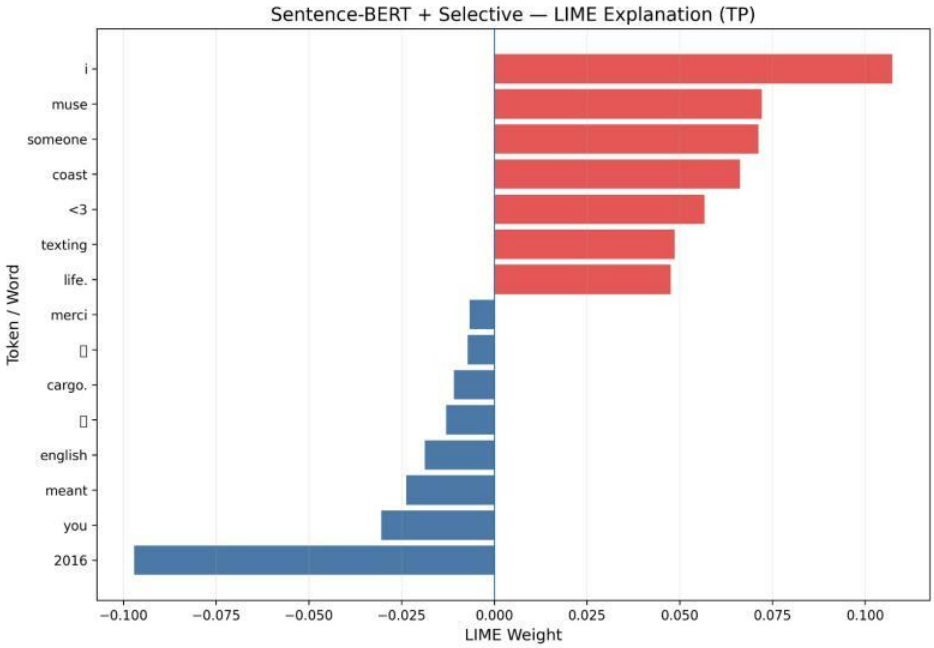


Figure 10: LIME explanation for a true-positive prediction by Sentence-BERT + Selective

6.6.4 Explainable AI Coverage

The t-SNE, SHAP and LIME analyses were produced for selected model–condition combinations and are presented as case studies of the representation structure, metadata contribution and an individual prediction. They are not evidence that every technique was applied uniformly to all 28 configurations, and they supplement, not replace, the accuracy, precision, recall, F1-score and ROC-AUC results.

6.7 Comparison with the Base Study

The present study follows the user-level depression-detection setting of the base study [4] but examines a wider range of architectures and representation conditions. Table 8 compares the best configuration of each study. The comparison lists the reported numbers only and is not a replication.

Table 8: Best configuration of the present study and of the base study [4]

Study	Configuration	Input length	Accuracy (%)	F1-score (%)	ROC-AUC (%)
Present study	Sentence-BERT + Raw	512 tokens	92.96	89.49	97.44
Base study [4]	Longformer + Selective + metadata	4096 tokens	90.36	87.15	96.67

The best configuration of the present study has an accuracy 2.60 percentage points higher, an F1-score 2.34 points higher and a ROC-AUC 0.77 points higher than the best configuration reported in the base study. The experimental designs differ, however: the base study used a 4096-token Longformer with Selective Cleaning, whereas this study evaluates seven architectures under four conditions with a fixed 512-token input. The studies also differ in the architecture sets and experimental matrices, and they use different numbers of metadata features (53 here). The differences in the numbers are therefore a comparison of reported configurations and not evidence that one architectural or preprocessing choice caused the difference; the effect of context length remains for future controlled experiments.

6.8 Discussion of the Research Questions

RQ1: How does the representation affect performance? The representation affects performance, but not in the same way for each architecture. Raw representation gave the best F1-score for four architectures (EmoBERT, MentalBERT, Sentence-BERT, BigBird), Emotion-Based representation for two (BERT, DistilBERT) and Selective representation for one (Longformer). Aggressive Cleaning was not best for any architecture, so more extensive cleaning is not a general way to improve performance.

RQ2: How does the architecture affect performance? Each of the seven architectures has its own profile. Sentence-BERT + Raw had the highest F1-score (89.49%) and accuracy (92.96%), MentalBERT + Raw followed (89.28%), and Longformer + Selective had the highest ROC-AUC (97.89%). The differences depend on the metric and on the condition.

RQ3: Does the effect of the representation depend on the architecture? Yes. The best condition differs between architectures, and the effect of Selective versus Aggressive cleaning also varies. Preprocessing and representation should be evaluated together with the downstream architecture, not assumed to act equally on all models.

RQ4: How do long-context architectures compare with conventional ones? Longformer and BigBird were competitive, but the results do not show that long-context architectures are generally better, since Sentence-BERT + Raw and MentalBERT + Raw achieved higher F1-scores than Longformer + Selective. Because all 28 configurations use 512 tokens, the study does not show whether extended context capacity improves performance; that would need controlled comparisons with different input lengths.

6.9 Discussion Synthesis

The way text is. How the transformer structure functions together play a key role in identifying depression at the user level. There wasn't a best way to represent text across all 28 different setups. Sentence-BERT with data achieved the highest F1-score at 89.49%. Longformer with data had the highest ROC-AUC at 97.89%. MentalBERT using data also performed well with an F1-score of 89.28% and a ROC-AUC of 97.68%. The raw method of text representation was the best for four models while the emotion-based approach worked best for two models. This shows that certain representations can help models but heavily cleaning the data did not consistently improve results.

These findings build on the base study [4] by testing seven models across four text representation methods using 512 tokens. The setups vary from one another. The results support the idea that the model and text representation work together at this input length and with this kind of preprocessing. However the results do not explain the impact of context length, model type or preprocessing. The choice of model for identifying depression at the user level should depend on how the model and text representation work as well as which performance metric is most important. The numbers reported are classification results from the tested dataset. They are not diagnoses and do not prove that any particular feature, in the data causes depression.

Chapter 7
Conclusion and Future Work

7.1 Conclusion

This project created a controlled system for identifying depression at the user level using media timelines. It used transformer-based language models and user-level metadata. The study examined text representations and model structures. It used four methods of representing text—Selective and Emotion-Based—and seven transformer models—BERT, DistilBERT Sentence-BERT, MentalBERT, EmoBERT, Longformer and BigBird. This produced 28 combinations arranged in a 4 by 7 table.

The framework had features: removing anchors using 53 user-level metadata features applying gated multimodal fusion and following a fixed user-level train, validation and test split. These steps helped keep the evaluation process controlled.

The performance of a text representation depended a lot on the transformer model used. Raw text worked best with Sentence-BERT, MentalBERT, EmoBERT and BigBird. Emotion-Based text worked best with BERT and DistilBERT. Selective text worked best with Longformer. The best setup was Sentence-BERT with text, which got an F1-score of 89.49 % accuracy of 92.96 % and ROC-AUC of 97.44 %. MentalBERT with text reached an accuracy of 92.83 % and an F1-score of 89.28 %. Longformer with text had the ROC-AUC at 97.89 %. A slightly lower F1-score of 88.72 %. Aggressive cleaning did not help any of the seven models. Stronger normalization did not consistently improve results.

Longformer and BigBird were competitive but not clearly better overall. All models used an input limit so the study does not show how longer inputs, like the 4096-token Longformer in the base study would affect performance. That result used a setup so it cannot be compared directly to show the effect of context length.

The diagnostic-anchor control, fixed user-level splits training-metadata normalization and training-only class balancing created a basis, for comparison. However these steps do not completely stop the risk of data leakage or bias. The t-SNE, SHAP and LIME analyses were used to explore how the representations worked how metadata influenced

them and how predictions were made for users. These tools were not meant to give explanations or prove causation. They were only meant to show patterns and support understanding.

7.2 Limitations

- Experiments are carried out using a single social media dataset at the user level, hence making the results hard to generalize to other datasets, as cross-population generalization is a problem in depression detection [31].
- The experimental set-up varies in terms of architecture and representation but not the input size, hence there is no way to determine different sizes of context.
- Differences are brought about by various pre-processing procedures, hence making it hard to determine the exact pre-processing procedure that leads to differences.
- Explanation of configurations was carried out in some of the configurations but not in all the 28 configurations.
- No significance test was performed on the experiment, and the differences between configurations are descriptive.

7.3 Future Work

- Comparisons among platforms and cultures for the purpose of verifying whether these combinations are valid not only for the given dataset but also for other datasets.
- Experimentation with different input sizes such as 512 and 4096 words with the other variables being held constant.
- Analysis using bigger histories and different languages with time or hierarchically.
- Other forms of representation for emotions or mental state, other fusion approaches, other context selection strategies.
- More thorough explanation and robustness analysis, including contribution consistency and false positive/negative analysis.

7.4 Ethical Aspects

The categories of the benchmark dataset are based on self-report of the subjects on social media; therefore, the problem statement is about spotting signs of depression through language and behavior, and not making diagnosis of it. Similarly to the case of the base

study [4], such models should only be viewed as pre-screening or risk detection tools, and their outputs should not affect the professional decision-making process. The results of the SHAP and LIME approaches relate to the functioning of the model on test data, and they are neither clinical nor causal in their nature. Because of the specific nature of the dataset used (single platform, lack of sub-group analysis), there might be some bias in the functioning of the model on various sub-groups; cross-population evaluation [31] needs to be conducted before using the model in practice.

References

1. Sen, G., Jia, J., Nie, L., Feng, F., Zhang, C., Hu, T., Chua, T.-S., Zhu, W.: Depression Detection via Harvesting Social Media: A Multimodal Dictionary Learning Solution. (2017).
2. Garg, M.: Mental Health Analysis in Social Media Posts: A Survey. Archives of Computational Methods in Engineering. 30, 1819–1842 (2023). https://doi.org/10.1007/s11831-022-09863-z.
3. Arshad, I., Naseem, S., Mussarat, M., Zanib, A., Al-Otaibi, S., Yousif, A., Mahmood, T., Khan, A.R.: A Multi-Platform Approach to Optimizing Depression Detection in Social Media Through Feature Engineering and Contextual Analysis. International Journal of Computational Intelligence Systems. 19, (2026). https://doi.org/10.1007/s44196-026-01225-y.
4. Can, H., Yücedag, I.: User-Level Depression Detection Using Long-Context Transformers and Behavioral Data. IEEE Access. 14, 73621–73645 (2026). https://doi.org/10.1109/ACCESS.2026.3689467.
5. Chen, C., Li, F., Chen, C., Chen, H., Lin, Y.: Heterogeneous subgraph network with prompt learning for interpretable depression detection on social media. Knowl.-Based Syst. 315, (2025). https://doi.org/10.1016/j.knosys.2025.113215.
6. Shah, S.M., Gillani, S.A., Baig, M.S.A., Saleem, M.A., Siddiqui, M.H.: Advancing depression detection on social media platforms through fine-tuned large language models. Online Social Netw. Media. 46, (2025). https://doi.org/10.1016/j.osnem.2025.100311.
7. Abdelkadir, N.A., Zhang, C., Mayo, N., Chancellor, S.: Diverse perspectives, divergent models: Cross-cultural evaluation of depression detection on Twitter. In: Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Human Lang. Technol. pp. 672–680 (2024). https://doi.org/10.18653/v1/2024.naacl-short.58.
8. Zogan, H., Razzak, I., Jameel, S., Xu, G.: Hierarchical convolutional attention network for depression detection on social media and its impact during pandemic. IEEE J. Biomed. Health Informat. 28, 1815–1823 (2024). https://doi.org/10.1109/JBHI.2023.3243249.
9. Zogan, H., Razzak, I., Jameel, S., Xu, G.: DepressionNet: Learning multi-modalities with user post summarization for depression detection on social media. In: Proc. 44th Int. ACM SIGIR Conf. Res. Develop. Inf. Retr. pp. 133–142 (2021). https://doi.org/10.1145/3404835.3462938.
10. Li, Z., An, Z., Cheng, W., Zhou, J., Zheng, F., Hu, B.: MHA: A multimodal hierarchical attention model for depression detection in social media. Health Inf. Sci. Syst. 11, (2023). https://doi.org/10.1007/s13755-022-00197-5.
11. Cui, B., Wang, J., Lin, H., Zhang, Y., Yang, L., Xu, B.: Emotion-based reinforcement attention network for depression detection on social media: Algorithm development and validation. JMIR Med. Informat. 10, (2022). https://doi.org/10.2196/37818.
12. Burdisso, S.G., Errecalde, M., Montes-y-Gómez, M.: t-SS3: a text classifier with dynamic n-grams for early risk detection over text streams. (2020). https://doi.org/10.1016/j.patrec.2020.07.001.
13. Jones, G.J.F., Lawless, S., Gonzalo, J., Kelly, L., Goeuriot, L., Mandl, T., Cappellato, L., Ferro, N. eds: Experimental IR Meets Multilinguality, Multimodality, and Interaction. Springer International Publishing, Cham (2017). https://doi.org/10.1007/978-3-319-65813-1.

14. Deng, S., Cheng, X., Hu, R.: Detecting depression and its severity based on social media digital cues. Ind. Manage. Data Syst. 123, 3038–3052 (2023). https://doi.org/10.1108/imds-12-2022-0754.

15. Stankevich, M., Smirnov, I., Kiselnikova, N., Ushakova, A.: Depression detection from social media profiles. In: Proc. Int. Conf. Data Anal. Manag. Data Intensive Domains (DAMDID). pp. 181–194. Springer, Cham, Switzerland (2019). https://doi.org/10.1007/978-3-030-51913-1_12.

16. Aduwamai Gadzama, W., Gabi, D., Sule Argungu, M., Umar Suru, H.: Hybrid BERT-GRU approach for depression detection on social media post. BIMA J. Sci. Technol. GOMBE. 9, 92–109 (2025). https://doi.org/10.64290/bima.v9i1a.888.

17. Haryoko, P., Syukur, A., Rijati, N.: Development of a mental health classifier using LSTM and text preprocessing techniques. Scientific J. Informat. 12, 77–86 (2025). https://doi.org/10.15294/sji.v12i1.21216.

18. Devlin, J., Chang, M.-W., Lee, K., Toutanova, K.: BERT: Pre-training of deep bidirectional transformers for language understanding. In: Proc. NAACL-HLT. pp. 4171–4186 (2019). https://doi.org/10.18653/v1/N19-1423.

19. Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., Stoyanov, V.: RoBERTa: A Robustly Optimized BERT Pretraining Approach. (2019).

20. Sanh, V., Debut, L., Chaumond, J., Wolf, T.: DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. (2020).

21. Ji, S., Zhang, T., Ansari, L., Fu, J., Tiwari, P., Cambria, E.: MentalBERT: Publicly Available Pretrained Language Models for Mental Healthcare. (2022).

22. Poswiata, R., Perelkiewicz, M.: OPI@LT-EDI-ACL2022: Detecting signs of depression from social media text using RoBERTa pre-trained language models. In: Proc. 2nd Workshop Lang. Technol. Equality, Diversity Inclusion. pp. 276–282 (2022). https://doi.org/10.18653/v1/2022.ltedi-1.40.

23. Sun, H., Liu, J., Chai, S., Qiu, Z., Lin, L., Huang, X., Chen, Y.: Multi-modal adaptive fusion transformer network for the estimation of depression level. Sensors. 21, 4764 (2021). https://doi.org/10.3390/s21144764.

24. Sharma, A., Parmar, M.: A survey on text pre-processing and feature extraction techniques for sentiment analysis of Twitter data. Int. Res. J. Comput. Sci. 8, 271–278 (2021).

25. Camacho-Collados, J., Pilehvar, M.T.: On the role of text preprocessing in neural network architectures: An evaluation study on text categorization and sentiment analysis. In: Proc. EMNLP Workshop BlackboxNLP: Analyzing Interpreting Neural Netw. NLP. pp. 40–46 (2018). https://doi.org/10.18653/v1/w18-5406.

26. Alzahrani, E., Jololian, L.: How different text-preprocessing techniques using the BERT model affect the gender profiling of authors. In: Proc. Adv. Mach. Learn. pp. 1–8 (2021). https://doi.org/10.5121/csit.2021.111501.

27. Guo, Y., Ge, Y., Yang, Y.-C., Al-Garadi, M., Sarker, A.: Comparison of pretraining models and strategies for health-related social media text classification. Healthcare. 10, (2022). https://doi.org/10.3390/healthcare10081478.

28. Lundberg, S.M., Lee, S.-I.: A unified approach to interpreting model predictions. In: Proc. Adv. Neural Inf. Process. Syst. pp. 4765–4774 (2017).

29. Han, S., Mao, R., Cambria, E.: Hierarchical Attention Network for Explainable Depression Detection on Twitter Aided by Metaphor Concept Mappings. (2022).

30. Beltagy, I., Peters, M.E., Cohan, A.: Longformer: The Long-Document Transformer. (2020).

31. Zaheer, M., Guruganesh, G., Dubey, K.A., Ainslie, J., Alberti, C., Ontanon, S., Pham, P., Ravula, A., Wang, Q., Yang, L., Ahmed, A.: Big bird: Transformers for longer sequences. In: Proc. Adv. Neural Inf. Process. Syst. pp. 17283–17297 (2020).

32. Reimers, N., Gurevych, I.: Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks.

33. Song, K., Tan, X., Qin, T., Lu, J., Liu, T.-Y.: MPNet: Masked and Permuted Pre-training for Language Understanding. (2020).

34. Ribeiro, M.T., Singh, S., Guestrin, C.: "Why should i trust you?" Explaining the predictions of any classifier. In: Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 1135–1144. Association for Computing Machinery (2016). https://doi.org/10.1145/2939672.2939778.

35. Van Der Maaten, L., Hinton, G.: Visualizing Data using t-SNE. (2008).